Software Test Plan (STP)

Project: Railway Ticket Reservation System

Version	1.0
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Status	Final
Related Document	Railway Ticket Reservation System SRS, Version 1.0

1. Introduction

Purpose: This document defines the test plan for the Railway Ticket Reservation System Version 1.0. It outlines the objectives, scope, test strategy, resources, schedule, responsibilities, and criteria for testing the system.

Scope: Testing covers train searching, train details and seat availability, ticket booking, passenger information validation, seat allocation, PNR generation, fare calculation, booking confirmation, booking retrieval, ticket cancellation, refund calculation, persistent booking records, and administrator functions.

References: Railway Ticket Reservation System SRS Version 1.0; Software Test Plan template for the Software Engineering project.

Definitions: SRS (Software Requirements Specification), STP (Software Test Plan), RTM (Requirements Traceability Matrix), PNR (Passenger Name Record), UAT (User Acceptance Testing).

2. Test Items

- Train Search module
- Train Details and Seat Availability module
- Ticket Booking module
- Passenger Information Validation
- Seat Allocation
- PNR and Fare Management
- Ticket Cancellation and Refund
- Booking Management and Persistent File Storage
- Administrator Authentication
- Administration and Train/Booking Data Management
- Menu-driven Console Interface

3. Features to be Tested

Features mapped to SRS requirement IDs:

Requirement ID	Feature	Test Case
RT-F001	Display all available trains	TC-TRAIN-01
RT-F002	Search trains using source and destination	TC-TRAIN-02
RT-F003	Search train using train number	TC-TRAIN-03
RT-F004	Display available seats	TC-TRAIN-04
RT-F005	Select train and enter passenger details	TC-BOOK-01
RT-F006	Validate passenger details	TC-BOOK-02
RT-F007	Check seat availability before booking	TC-BOOK-03
RT-F008	Assign an available seat	TC-BOOK-04

RT-F009	Generate unique PNR	TC-BOOK-05
RT-F010	Calculate and display ticket fare	TC-BOOK-06
RT-F011	Display booking confirmation	TC-BOOK-07
RT-F012	Search booking using PNR	TC-CAN-01
RT-F013	Display booking details before cancellation	TC-CAN-02
RT-F014	Cancel valid booking after confirmation	TC-CAN-03
RT-F015	Make cancelled seat available	TC-CAN-04
RT-F016	Update booking status to CANCELLED	TC-CAN-05
RT-F017	Calculate and display refund amount as 90% of fare (10% cancellation charge)	TC-CAN-06
RT-F018	View booking details using PNR	TC-MGT-01
RT-F019	Prevent duplicate cancellation	TC-MGT-02
RT-F020	Maintain persistent booking records	TC-MGT-03
RT-F021	Add new train information; reject duplicate train numbers	TC-ADMIN-01
RT-F022	Modify existing train information subject to booking/capacity rules; existing bookings retain stored fares	TC-ADMIN-02
RT-F023	View all train and booking records	TC-ADMIN-03
RT-F024	Require administrator authentication	TC-ADMIN-04
RT-F025	Delete a train only when it has no confirmed bookings	TC-ADMIN-05

Non-functional requirements:

Requirement ID	Feature	Test Case
RT-NF001	Response time ≤ 2 seconds for train search with up to 10,000 train records	TC-NFR-01
RT-NF002	Consistent seat availability after booking and cancellation	TC-NFR-02
RT-NF003	Simple menu-driven usability	TC-NFR-03
RT-NF004	Booking records remain available after restart	TC-NFR-04
RT-NF005	Logical component separation	TC-NFR-05

Security requirements:

Requirement ID	Feature	Test Case
PRJ-SR001	Validate user input	TC-SEC-01
PRJ-SR002	Reject invalid train numbers and PNRs	TC-SEC-02
PRJ-SR003	Prevent duplicate PNRs	TC-SEC-03
PRJ-SR004	Prevent booking with no seats	TC-SEC-04
PRJ-SR005	Require administrator authentication	TC-SEC-05

4. Features Not to be Tested

- Real railway reservation servers and live railway data.
- Payment gateway integration and real payment processing.
- Real-time railway APIs and external network communication.
- Centralized database functionality.
- Railway infrastructure and external hardware outside the application.

5. Test Approach / Strategy

5.1 Levels

- Unit tests (module-level).
- Integration tests between interacting modules.
- System tests covering end-to-end passenger and administrator functionality.
- Acceptance tests (UAT).

5.2 Types

- Functional testing (core features).
- Regression testing.
- Performance testing (response time and defined dataset size; search time measured after data is loaded into memory).
- Usability testing (menu clarity and ease of use).
- Reliability and persistence testing.

Architecture inspection (static review of module dependencies and layer boundaries) for TC-NFR-05.

Entry Criteria: Stable build delivered, test data available, and test environment ready.

Exit Criteria: 100% of planned test cases executed, no critical functional/security/data-persistence defects open, and all applicable acceptance criteria satisfied.

5.3 Security Validation

- Validate all user input before processing.
- Reject invalid train numbers and PNR values.
- Verify PNR uniqueness.
- Verify booking is prevented when no seats are available.
- Verify administrator authentication before administrative functions.

Verify duplicate train numbers are rejected.

Verify trains with confirmed bookings cannot be deleted.

Verify train capacity cannot be reduced below the number of confirmed bookings.

Verify cancellation refund equals 90% of the stored booking fare.

6. Test Environment

Hardware: Standard computer with keyboard, display, processor, and sufficient storage for application files. Performance measurements shall record the test machine, compiler/build configuration, dataset size, and measured search time.

Software: C++ application, Windows/Linux operating system, standard C++ compiler such as g++ or GCC, and C++ Standard Library.

Storage: Local text files used for persistent train and booking information.

Test Data: Simulated train records, passenger details, seat availability, fares, PNRs, administrator credentials, confirmed/cancelled bookings, and a dataset of up to 10,000 train records. The 10,000-record performance dataset shall contain valid train records with known source/destination values and shall be reused for repeatable runs.

TC-NFR-01 measurement rule: load the dataset before timing, execute the defined search repeatedly, and require each measured search to complete within 2 seconds; startup and initial file loading time are excluded.

Tools: GitHub repository and issue tracking for source control and defect reporting.

7. Test Schedule

Milestone	Date / Period	Responsibility
Test case design	Before test execution	QA / Test Team
Test environment setup	Before test execution	Development Team

Test execution	After stable build is available	Test Team
Regression testing	After defect fixes	Test Team
UAT	After system testing	Project Team

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Mohammed Farhan Ahmed	Prepare plan, coordinate execution
Test Engineer	Monisha Agarwal	Design & execute test cases, log defects
Developer	Nandita N Patil	Support defect fixes and triage
Product Owner	Lakshmi Pranathi Tummalapenta	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early test builds and prepare test cases in parallel.
Test environment downtime	Maintain a working backup development/test environment.
Incorrect or corrupted local test data	Maintain controlled test files and restore known test data when required.
Integration defects between modules	Perform unit and integration testing before full system testing.
Requirement or implementation changes	Run impacted regression tests after changes.

11. Assumptions & Dependencies

- The C++ compiler and required execution environment will be available.
- Test data will be prepared before test execution.
- Local text files will be available for persistent storage.
- The system will operate without external network communication.
- A stable build will be provided for system and acceptance testing.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment is unavailable.
- The build is too unstable to execute planned tests.
- Blocking defects prevent execution of a significant portion of the test cases.

Resume testing if:

- Blocking defects are resolved.

- The environment is stabilized.
- A stable build is available.

13. Test Case Management & Traceability

RTM: The Requirements Traceability Matrix maps SRS requirements to test cases. The following mapping is used for test execution and coverage. Requirement and test IDs use the standardized formats RT-F001, RT-NF001, PRJ-SR001 and TC-SEC-01/TC-NFR-01/TC-ADMIN-01.

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PRJ-SR003	Prevent duplicate PNRs	TC-SEC-03
PRJ-SR004	Prevent booking with no seats	TC-SEC-04
PRJ-SR005	Require administrator authentication	TC-SEC-05

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage

Reports:

- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Mohammed Farhan Ahmed	
Developer	Nandita N Patil	
Test Engineer	Monisha Agarwal	
Product Owner	Lakshmi Pranathi Tummalapenta	